IAN MERRIMAN

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RELEVANT PROJECTS

Thrust-Vectoring Rocket Project – Flight Controls Team *July 2025 – Present*

- Developed and implemented a 6-DOF flight dynamics simulation in MATLAB/Simulink to model thrust-vectoring vehicle behavior and evaluate closed-loop control performance.
- Performed simulation-based verification of system behavior against expected stability criteria.

Project BlueSpear – AIAA at UTA *July 2024 – Present*

- Co-authored an approved proposal for a 100 km rocket (Kármán line).
- Contributed to system-level trade studies across trajectory, propulsion, and safety disciplines to support vehicle feasibility and mission architecture development.

Rocketry Projects – Aero Mavs *Sept 2022 – Present*

- Designed, built, and flown three personal rockets, including a nose cone with an automatic drogue parachute system.
- Manufacturing experience with body tubes, fin attachments, and integration of recovery systems.
- Member of the Rocketry Division contributing to competition-grade vehicle development.

Self-Balancing Table Project – Controls Programmer *Jan 2025 – May 2025*

- Implemented a closed-loop feedback control system on a Raspberry Pi Pico to stabilize a ball using servo-actuated tilting surfaces.
- Programmed real-time control logic in MicroPython to maintain system stability.
- Tuned control parameters to minimize oscillation and response time.

LEADERSHIP & ORGANIZATIONS

Aero Mavs – UTA Aerospace Student Organization

President | July 2024 – July 2025

- Oversaw operations, managed cross-disciplinary technical divisions, and coordinated projects and competitions.
- Secured $4,000+ in funding and doubled membership from 53 to 110.
- Directed both technical and administrative teams to meet organizational goals.

AIAA at UTA

Navigation Lead | Feb 2026 – Present

- Lead development of trajectory analysis and flight path planning for a Kármán-line rocket.
- Defining performance requirements for trajectory and navigation subsystems.

TECHNICAL SKILLS

- **Software & Tools:** MATLAB/Simulink, C++, SolidWorks, Raspberry Pi Pico, MicroPython, GitHub
- **Engineering:** Flight Dynamics, Spacecraft Dynamics, Automatic Controls, Orbital Mechanics, Control System Analysis, Aerospace Software Design
- **Certifications:** L1 High-Power Rocketry Certification (Tripoli Rocketry Association)
- **Other:** 6-DOF Simulation, Trajectory Modeling, Control System Design, Numerical Methods

EDUCATION

University of Texas at Arlington Exp. August 2026

Bachelor of Science, Aerospace Engineering GPA 3.6

• Dean's List, Fall 2023 • Dean's List, Spring 2024